DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
Faculty of Computing & Applied Sciences
University of Southern Punjab, Multan

Web-Based Centralized Platform That Optimizes Food Waste Redistribution by Linking Restauran Registered Charitable Organizations

FINAL YEAR PROJECT (FYP) · BSCS SESSION 2024–2026 · SHAREMEAL ECOSYSTEM

DESCRIPTION

Food wastage is one of the most critical socio-economic and environmental paradoxes in Pakistan. While millions suffer severe nutritional deprivation, banquet halls, hotels, and restaurants discard tons of wholesome surplus food daily due to the lack of an organized redistribution network. The **ShareMeal Platform** is an enterprise-grade web-based centralized system built on **ASP.NET Core 8.0, C# 12, Entity Framework Core, and Google Gemini AI**. The system establishes an instant digital bridge linking surplus food donors directly with certified charitable NGOs. By digitizing inventory, automating dispatch logistics, enforcing strict hygiene and expiry SOPs, and providing voice-enabled AI assistance, ShareMeal permanently eliminates food waste and feeds impoverished communities sustainably.

COMPARATIVE ANALYSIS

Operational Feature	Manual Calling	WhatsApp Groups	ShareM
Real-Time Matching	No	Partial	✔ Yes (I
Double-Claim Prevention	No	No (High Chaos)	✔ Yes (A
Verified NGO Authentication	No	No	✔ Yes (A
Food Safety & Expiry Tracking	No	No	✔ Yes (A
24/7 AI-Powered Assistance	No	No	✔ Yes (G MealBot
Voice Accessibility (Bilingual)	No	No	✔ Yes (V
Live GPS Route Mapping	No	Manual Links	✔ Yes (L
Audit Trail & Impact Reports	No	No	✔ Yes (A PDF/Log

SCOPE OF THE SYSTEM

- **Nationwide Web Accessibility:** Responsive cloud architecture operating seamlessly across desktop, tablet, and mobile browsers without requiring native app installs.
- **Multi-Tier Role Management:** Role-segregated secure portals for Food Donors (Restaurants), Claiming Charities (NGOs), and System Administrators.
- **Real-Time Surplus Food Matching:** Dynamic listings specifying portions (servings/kg), cooking timestamp, thermal state, and expiry countdown.
- **Atomic Concurrency Claim Engine:** Database-level transaction locking preventing duplicate claims or race conditions between competing charities.
- **Bilingual Multimodal Gemini AI (MealBot):** 24/7 intelligent voice-enabled assistant offering automated food safety SOPs, navigation, and community guidance.
- **Interactive Geolocation Mapping:** Integration with Leaflet.js and GPS routing for pinpointing donor addresses, route estimation, and distribution verification.

RELATIONAL SCHEMA DIAGRAM

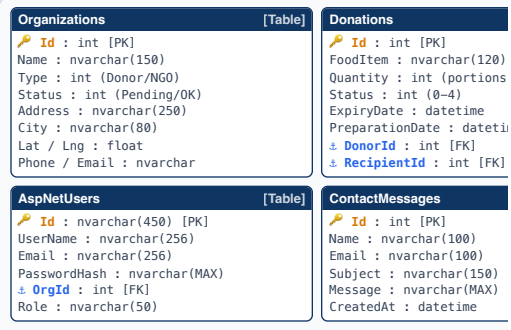


Figure: Normalized Relational Entity Relationship Diagram (1:M Referenti

FUNCTIONAL REQUIREMENTS

- **User Identity & Access:** Secure user registration, authentication, and role assignment (ASP.NET Core Identity).
- **Restaurant / Donor Module:** Create, manage, edit, and track surplus food listings with expiry countdown.
- **Charity / NGO Module:** Live marketplace feed, geolocation filtering, and one-click instant reservation.
- **Strict Verification Pipeline:** Admin verification workflows verifying government NGO credentials and NTN.
- **Status Lifecycle Engine:** Automated transitions across *Available* → *Claimed* → *In-Transit* → *Delivered*.
- **Food Safety & Hygiene SOPs:** Pre-claim verification of food cooking time and storage temperature.
- **MealBot AI Integration:** Natural language processing in Roman Urdu, Urdu, and English with Web Speech API.
- **System Analytics & Reporting:** Comprehensive PDF and visual telemetry for meals saved and carbon offset.

TOOLS & TECHNOLOGIES

METHODOLOGY (AGILE RAD MODEL)

The project adopted the **Rapid Application Development (RAD)** methodology under an Agile iterative lifecycle, emphasizing fast modular development, immediate stakeholder testing, and continuous deployment:

1. Requirements & Analysis
2. Architecture & Schema Design
3. Iterative Prototyping
4. QA & AI Integration
5. Cloud Deployment

Hardware Tools

- **Processor:** Intel Core i7 / i5 Multi-Core
- **Memory:** 16GB DDR4 High-Speed RAM
- **Storage:** 512GB NVMe SSD
- **Display:** 1080p FHD / 4K Workstation
- **Testbed:** iOS Safari & Android Chrome

Software Tools

- **Backend:** ASP.NET Core
- **ORM & DB:** Entity Framev
- **Frontend:** Modern HTML5
- **AI Engine:** Google Gemin
- **Maps & Audio:** Leaflet & V
- **Cloud / DevOps:** Docker

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